

Qatar-1b

R-1249

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_251014 · created 2026-07-31T11:45:28+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460961.3674 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.041
Transit depth (Rp/Rs)^2	3.7 +/- 1.57 [%]
Orbital Inclination (inc)	89.06 +/- 2.17
Airmass coefficient 1 (a1)	0.834 +/- 0.029
Airmass coefficient 2 (a2)	0.13 +/- 0.1
Scatter in the residuals of the lightcurve fit is	3.54 %
Best Comparison Star	#3 - [156, 289]
Optimal Aperture	3.28
Optimal Annulus	12.57
Transit Duration (day)	0.0327 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.041 vs 0.1463 (deviation 1.09 σ)
Duration fit vs published	0.0327 d vs 0.0692 d (deviation 9.1 σ)
Mid-time O-C	-2040.7 min
Depth SNR	4.6
Residual scatter	3.54 %

Light curve

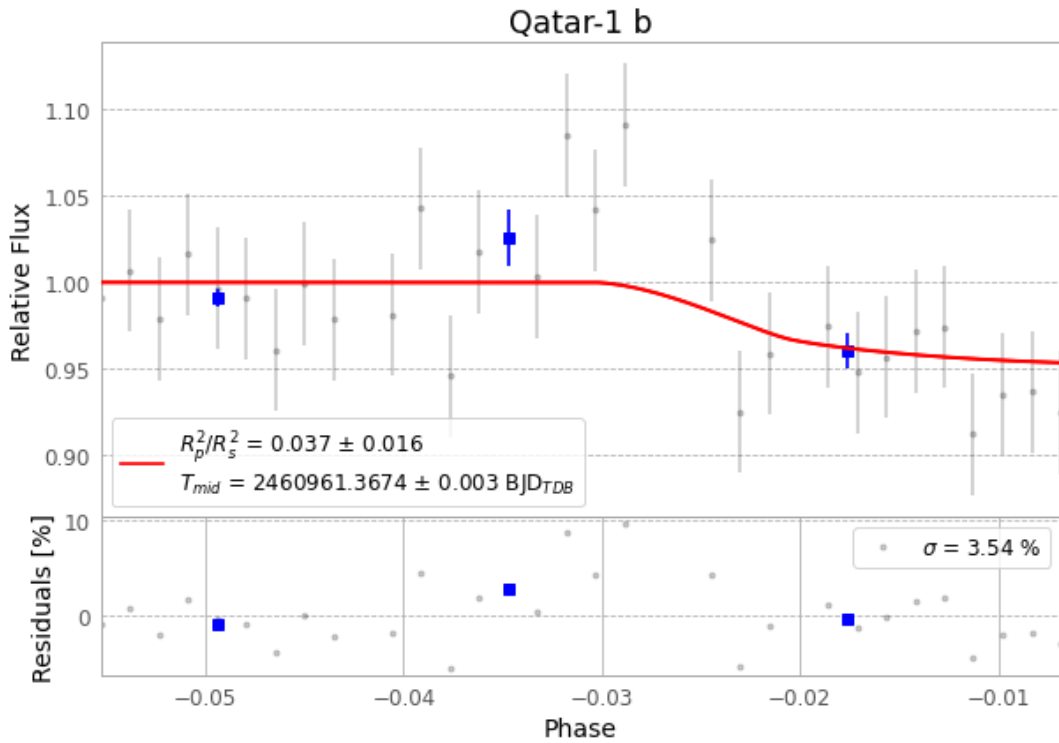


Figure 1 — FinalLightCurve_Qatar-1 b_2025-10-13.png (SHA-256 559dd7cd42fbc41d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [253, 361], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a90e6de4ebae5e1d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

30 FITS frames (20 MB), Qatar-1251014045716.FITS → Qatar-1251014063616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-251014.log (4128fd559639...), FinalLightCurve_Qatar-1 b_2025-10-13.png (559dd7cd42fb...), FinalLightCurve_Qatar-1 b_2025-10-13.csv (858c3c440285...)

Compute resources

Wall time 355.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:45Z.